The Æther-flow Interpretation of Relativity

Primitive-Reservoir Observer-Reduction
Combined Response Substrate Law
Necessity / Uniqueness from Zero-Hidden Selector Generator Law

Abstract

The current benchmark-facing theorem on the active primitive-reservoir observer-reduction line already sharpened the burden once more: the completion-balance ambient dynamics package is no longer merely available, because it is derived canonically from a deeper combined response substrate law and is necessary there up to the obvious hidden-factor identification. The remaining honest task is therefore no longer to justify the ambient package. It is to justify why the combined response substrate law itself should hold.

The present note makes that next move in a form aligned with the live routing safeguard. For each sector it starts from a deeper zero-hidden selector generator law consisting of a compact convex selector-state body, affine observer-data and shell-response readouts, a self-adjoint idempotent hidden selector, an exact selector-level body splitting over selector-zero and hidden directions, a positively gapped hidden-sector operator on the selector image, symmetry and time-reversal actions preserving the whole selector package, an exact zero-response shell realization on the selector-zero locus, fixed-data shell solvability on that same locus, and a coercive selector-law inequality on data-invisible directions modulo the inert zero-hidden kernel.

From that selector law one derives canonically the currently live combined response substrate law itself. The primitive substrate state is the faithful quotient of the selector-zero locus by the inert zero-hidden kernel, the primitive substrate body is the quotient body crossed with the hidden selector body, the affine combined response map is the descended selector data and shell readout together with the explicit hidden coordinate, the response image factorizes exactly over the hidden body, the positive hidden-sector operator is inherited unchanged, the symmetry and time-reversal structure descend, the exact zero-response shell realization survives on the faithful quotient, fixed-data shell solvability descends, and the response-kernel coercive inequality is recovered with no remaining inert zero-hidden freedom. Within this selector-law class the combined response substrate law is not merely available. It is necessary and unique up to the harmless affine choice of faithful zero-hidden origin and the obvious orthogonal identifications on the faithful quotient and hidden factor.

The benchmark boundary remains fixed throughout. The one operative metric is still exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, the low-energy matter law is not enlarged, and stronger first-principles promotion language is still not licensed. The gain is therefore disciplined and benchmark-facing: the current combined response stopping point is no longer left as an independently inserted package, but is forced by a deeper explicit selector law after quotienting out exactly the selector-zero directions already invisible to the recorded data and shell readouts.

1. Introduction

The active derivational program of Æther-flow Interpretation of Relativity has reached a new upstream burden. The primitive-observer-reduction datum theorem, the defect-fiber factorization theorem, the one-metric output theorem, the chain-forcing theorem, the selector theorem, the stationary-construction theorem, the explicit-package theorem, the primitive-normal-form theorem, the ambient-shell-law theorem, the combined-response theorem, the later completion-core and trace-core compressions, and the current ambient-dynamics forcing theorem have already discharged what was honestly open at their respective layers. The latest result matters because it changes what now counts as front-line progress. Once the completion-balance ambient dynamics package is itself forced by the deeper combined response substrate law, another ambient-package restatement, trace-core restatement, completion-core restatement, exact-shell affine nucleus restatement, or same-law combined-response relay no longer counts as the honest next move.

The remaining honest burden is therefore one level below the current combined-response frontier. The task is now to explain why the combined response substrate law itself should hold, rather than letting the compact primitive substrate body, the affine combined response map, the response-image factorization, the hidden-sector gap operator, the shell realization, the solvability statement, and the response-kernel coercivity mechanism stand as the new deepest disciplined assumptions. That is the burden addressed here.

The present note does not reopen the benchmark effective package. It does not alter the one operative completed metric, it does not introduce a second operative metric, and it does not enlarge the low-energy matter law. Its gain is narrower and more upstream. It shows that the current combined response substrate law is already a canonical and necessary reduction of a deeper zero-hidden selector generator law once the selector-zero directions invisible to the recorded data and shell readouts are quotiented out faithfully.

Framework and claim boundary. Æther / Æther-flow framework denotes the broader Æther / Æther-flow framework built on the claims that the Æther is the underlying four-dimensional substrate of reality, the Æther-flow is its intrinsic ordered motion, observed three-dimensional space is the local experiential slice of that deeper substrate, S-time is the experienced order of change arising from matter, light, and the Æther-flow, and observed expansion is the three-dimensional appearance of deeper four-dimensional ordered motion. Within that framework, Æther-flow Interpretation of Relativity denotes the exact-closure theory in which the effective gravitational dynamics are adopted to be exactly Einsteinian with universal matter coupling. In the active sequence, *adoption* means use of that established relativistic dynamics without claiming substrate derivation, while *derivation* is reserved for a first-principles recovery from explicit substrate variables.

The present note stays strictly inside that boundary. It does not claim that final substrate microphysics has already been found. Its narrower theorem is only that, on the active primitive-observer-reduction line, the combined response substrate law is itself a forced and unique output of a deeper zero-hidden selector generator law after faithful inert-kernel reduction.

2. Fixed Primitive Continuation Data

Definition 1 (Recorded primitive continuation data)**.** *Fix the following continuation data from the active primitive-reservoir observer-reduction line.*

1. *For each sector label* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*, a primitive forcing shell*

$$\mathcal{I}_\sigma^{\mathrm{for}},$$

a visible reduction map

$$\mathcal{R}_\sigma : \mathcal{I}_\sigma^{\mathrm{for}} \to X_\sigma,$$

and shell-level current and equilibrium carriers

$$\mathfrak{C}_\sigma^{\mathrm{cur}} : \mathcal{I}_\sigma^{\mathrm{for}} \to Z_\sigma^{\mathrm{cur}}, \qquad \mathfrak{C}_\sigma^{\mathrm{eq}} : \mathcal{I}_\sigma^{\mathrm{for}} \to Z_\sigma^{\mathrm{eq}}.$$

2. *The product shell*

$$\mathcal{S}_{\mathrm{prim}} \equiv \mathcal{I}_{\mathrm{car}}^{\mathrm{for}} \times \mathcal{I}_{\mathrm{cur}}^{\mathrm{for}} \times \mathcal{I}_{\mathrm{hom}}^{\mathrm{for}}.$$

3. *Fixed shell extractors*

$$\mathscr{E}_\sigma^{\mathrm{cur}} : Z_\sigma^{\mathrm{cur}} \to Y_\sigma^{\mathrm{cur}}, \qquad \mathscr{E}_\sigma^{\mathrm{eq}} : Z_\sigma^{\mathrm{eq}} \to Y_\sigma^{\mathrm{eq}},$$

together with the recorded shell composites

$$\mathcal{J}_\sigma^{\mathrm{prim}} = \mathscr{E}_\sigma^{\mathrm{cur}} \circ \mathfrak{C}_\sigma^{\mathrm{cur}}, \qquad \Theta_\sigma^{\mathrm{prim}} = \mathscr{E}_\sigma^{\mathrm{eq}} \circ \mathfrak{C}_\sigma^{\mathrm{eq}}. \tag{1}$$

4. *The recorded metric and ordinary-matter outputs*

$$\mathscr{G}_{\mathrm{out}} : \mathcal{S}_{\mathrm{prim}} \to \{\textit{observer metrics}\}, \qquad \mathscr{T}_{\mathrm{out}} : \mathcal{S}_{\mathrm{prim}} \times \Psi \to \{\textit{observer stress tensors}\},$$

satisfying

$$\mathscr{G}_{\mathrm{out}}(\mathbf{w}) = g^\sharp, \qquad \mathscr{T}_{\mathrm{out}}(\mathbf{w}, \psi) = T_{\mu\nu}^{\mathrm{matter}}[g^\sharp, \psi] \tag{2}$$

for every $\mathbf{w} \in \mathcal{S}_{\mathrm{prim}}$ *and every matter configuration* $\psi \in \Psi$*.*

Remark 1. *Definition 1 is not reopened here. The primitive shell data and the benchmark one-metric / ordinary-matter outputs remain fixed continuation data. The new burden begins one layer deeper again: why the combined response substrate law that now carries the active line should itself be forced.*

3. Target Combined Response Law

Definition 2 (Combined response substrate law with hidden-sector gap)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*. A* combined response substrate law with hidden-sector gap *for sector* σ *consists of the following data.*

1. *A finite-dimensional Euclidean primitive substrate space*

$$\mathcal{M}_\sigma$$

with a compact convex body

$$\mathcal{P}_\sigma \subset \mathcal{M}_\sigma.$$

2. *An observer-data target space*

$$\mathcal{Y}_\sigma^{\mathrm{obs}} \equiv \widehat{X}_\sigma \times Z_\sigma^{\mathrm{cur}} \times Z_\sigma^{\mathrm{eq}},$$

an ambient shell target

$$E_\sigma^{\mathrm{amb}},$$

a hidden response space

$$H_\sigma,$$

and the product response space

$$\mathcal{R}_\sigma \equiv \mathcal{Y}_\sigma^{\mathrm{obs}} \oplus E_\sigma^{\mathrm{amb}} \oplus H_\sigma.$$

3. *An affine combined response map*

$$\Xi_\sigma = (\Xi_\sigma^{\mathrm{dat}}, \Xi_\sigma^{\mathrm{sh}}, \Xi_\sigma^{\mathrm{hid}}) : \mathcal{M}_\sigma \to \mathcal{R}_\sigma.$$

4. *The response image body*

$$\mathcal{B}_\sigma^{\mathrm{resp}} \equiv \Xi_\sigma(\mathcal{P}_\sigma) \tag{3}$$

is compact and convex, and its shell slice

$$\mathcal{B}_\sigma^{\mathrm{resp}} \cap (\mathcal{Y}_\sigma^{\mathrm{obs}} \oplus E_\sigma^{\mathrm{amb}} \oplus \{0\})$$

has nonempty relative interior in

$$\mathcal{Y}_\sigma^{\mathrm{obs}} \oplus E_\sigma^{\mathrm{amb}} \oplus \{0\}.$$

5. *A compact convex hidden body*

$$D_\sigma \subset H_\sigma$$

with

$$0 \in \operatorname{int} D_\sigma,$$

such that the response image factorizes exactly as

$$\mathcal{B}_\sigma^{\mathrm{resp}} = \{(u, e, n) : (u, e, 0) \in \mathcal{B}_\sigma^{\mathrm{resp}}, \quad n \in D_\sigma\}. \tag{4}$$

6. *A positive self-adjoint hidden-sector operator*

$$K_\sigma^{\mathrm{hid}} : H_\sigma \to H_\sigma$$

with lower bound

$$\left\langle \eta, K_\sigma^{\mathrm{hid}} \eta \right\rangle \geq \kappa_\sigma^{\mathrm{hid}} \|\eta\|^2 \qquad \textit{for all } \eta \in H_\sigma \tag{5}$$

for some $\kappa_\sigma^{\mathrm{hid}} > 0$.

7. *A compact symmetry group*

$$\mathbb{H}_\sigma \subset O(\mathcal{Y}_\sigma^{\mathrm{obs}}) \times O(E_\sigma^{\mathrm{amb}}) \times O(H_\sigma)$$

and an orthogonal involution

$$\rho_\sigma \in O(\mathcal{Y}_\sigma^{\mathrm{obs}}) \times O(E_\sigma^{\mathrm{amb}}) \times O(H_\sigma)$$

preserving $\mathcal{B}_\sigma^{\mathrm{resp}}$*, the hidden-zero slice*

$$\mathcal{Y}_\sigma^{\mathrm{obs}} \oplus E_\sigma^{\mathrm{amb}} \oplus \{0\},$$

the zero shell value $0 \in E_\sigma^{\mathrm{amb}}$*, and* K_σ^{hid}*.*

8. *A smooth embedding*

$$\jmath_\sigma : X_\sigma \hookrightarrow \widehat{X}_\sigma$$

and a smooth shell realization map

$$\zeta_\sigma : \mathcal{I}_\sigma^{\mathrm{for}} \hookrightarrow \mathcal{M}_\sigma$$

such that

$$\zeta_\sigma(\mathcal{I}_\sigma^{\mathrm{for}}) \subset \mathcal{P}_\sigma \cap (\Xi_\sigma^{\mathrm{sh}})^{-1}(0) \cap (\Xi_\sigma^{\mathrm{hid}})^{-1}(0),$$

the map $\Xi_\sigma \circ \zeta_\sigma$ *is a diffeomorphism onto*

$$\mathcal{B}_\sigma^{\mathrm{resp}} \cap (\mathcal{Y}_\sigma^{\mathrm{obs}} \oplus \{0\} \oplus \{0\}),$$

and

$$\Xi_\sigma^{\mathrm{dat}}(\zeta_\sigma(w)) = (\jmath_\sigma(\mathcal{R}_\sigma(w)), \mathfrak{C}_\sigma^{\mathrm{cur}}(w), \mathfrak{C}_\sigma^{\mathrm{eq}}(w)) \tag{6}$$

for every $w \in \mathcal{I}_\sigma^{\mathrm{for}}$*.*

9. *Zero-response shell solvability on fixed data fibers:*

$$\mathrm{pr}_{\mathcal{Y}_\sigma^{\mathrm{obs}}}(\mathcal{B}_\sigma^{\mathrm{resp}}) = \mathrm{pr}_{\mathcal{Y}_\sigma^{\mathrm{obs}}}\Big(\mathcal{B}_\sigma^{\mathrm{resp}} \cap (\mathcal{Y}_\sigma^{\mathrm{obs}} \oplus \{0\} \oplus \{0\})\Big). \tag{7}$$

10. *Response-kernel coercivity on the primitive shell: there exists* $\kappa_\sigma^{\mathrm{resp}} > 0$ *such that for every*

$$w \in \mathcal{I}_\sigma^{\mathrm{for}}, \qquad \delta m \in T_{\zeta_\sigma(w)}\mathcal{M}_\sigma$$

satisfying

$$D\Xi_\sigma^{\mathrm{dat}}(\zeta_\sigma(w))\,\delta m = 0,$$

one has

$$\|D\Xi_\sigma^{\mathrm{sh}}(\zeta_\sigma(w))\,\delta m\|^2 + \Big\langle D\Xi_\sigma^{\mathrm{hid}}(\zeta_\sigma(w))\,\delta m, K_\sigma^{\mathrm{hid}} D\Xi_\sigma^{\mathrm{hid}}(\zeta_\sigma(w))\,\delta m\Big\rangle \geq \kappa_\sigma^{\mathrm{resp}}\,\|\delta m\|^2. \tag{8}$$

Remark 2. *Definition 2 is the target package whose origin is now at issue. The purpose here is not to restate it as another deeper stopping point, but to show that it is already a canonical output*

of a deeper zero-hidden selector generator law after faithful inert-kernel reduction.

4. Zero-Hidden Selector Generator Law

Definition 3 (Zero-hidden selector generator law)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*. A* zero-hidden selector generator law *for sector* σ *consists of the following data.*

1. *A finite-dimensional Euclidean selector-state space*

$$\mathcal{G}_\sigma^{\mathrm{sel}},$$

the observer-data space

$$\mathcal{Y}_\sigma^{\mathrm{obs}} \equiv \widehat{X}_\sigma \times Z_\sigma^{\mathrm{cur}} \times Z_\sigma^{\mathrm{eq}},$$

an ambient shell-response space

$$E_\sigma^{\mathrm{amb}},$$

and a compact convex selector-state body

$$\mathcal{B}_\sigma^{\mathrm{sel}} \subset \mathcal{G}_\sigma^{\mathrm{sel}}$$

with nonempty interior.

2. *An affine observer-data readout*

$$\Lambda_\sigma^{\mathrm{sel,dat}} : \mathcal{G}_\sigma^{\mathrm{sel}} \to \mathcal{Y}_\sigma^{\mathrm{obs}}$$

and an affine shell-response readout

$$\Lambda_\sigma^{\mathrm{sel,sh}} : \mathcal{G}_\sigma^{\mathrm{sel}} \to E_\sigma^{\mathrm{amb}}.$$

3. *A self-adjoint idempotent hidden selector*

$$\Sigma_\sigma^{\mathrm{hid}} : \mathcal{G}_\sigma^{\mathrm{sel}} \to \mathcal{G}_\sigma^{\mathrm{sel}}, \qquad {\Sigma_\sigma^{\mathrm{hid}}}^2 = \Sigma_\sigma^{\mathrm{hid}}, \qquad {\Sigma_\sigma^{\mathrm{hid}}}^* = \Sigma_\sigma^{\mathrm{hid}}.$$

Define the selector-zero locus

$$\mathcal{Z}_\sigma^0 \equiv \ker \Sigma_\sigma^{\mathrm{hid}},$$

its translation space

$$V_\sigma^0 \equiv \ker \Sigma_\sigma^{\mathrm{hid}},$$

and the hidden selector image

$$H_\sigma \equiv \mathrm{im}\, \Sigma_\sigma^{\mathrm{hid}}.$$

4. *A compact convex hidden body*

$$D_\sigma \subset H_\sigma$$

with

$$0 \in \mathrm{int}\, D_\sigma,$$

such that the selector-state body splits exactly as

$$\mathcal{B}_\sigma^{\mathrm{sel}} = \left\{ z + \eta : z \in \mathcal{B}_\sigma^{\mathrm{sel}} \cap \mathcal{Z}_\sigma^0, \quad \eta \in D_\sigma \right\}. \tag{9}$$

5. *A positive self-adjoint hidden-sector operator*

$$K_\sigma^{\mathrm{hid}} : H_\sigma \to H_\sigma$$

satisfying

$$\left\langle \eta, K_\sigma^{\mathrm{hid}} \eta \right\rangle \geq \kappa_\sigma^{\mathrm{hid}} \, \|\eta\|^2 \qquad \textit{for all } \eta \in H_\sigma \tag{10}$$

for some $\kappa_\sigma^{\mathrm{hid}} > 0$.

6. *A compact symmetry group*

$$\mathbb{S}_\sigma \subset O(\mathcal{G}_\sigma^{\mathrm{sel}}) \times O(\mathcal{Y}_\sigma^{\mathrm{obs}}) \times O(E_\sigma^{\mathrm{amb}})$$

and an orthogonal involution

$$\vartheta_\sigma^{\mathrm{sel}} \in O(\mathcal{G}_\sigma^{\mathrm{sel}}) \times O(\mathcal{Y}_\sigma^{\mathrm{obs}}) \times O(E_\sigma^{\mathrm{amb}})$$

preserving $\mathcal{B}_\sigma^{\mathrm{sel}}$, $\Lambda_\sigma^{\mathrm{sel,dat}}$, $\Lambda_\sigma^{\mathrm{sel,sh}}$, $\Sigma_\sigma^{\mathrm{hid}}$, D_σ, *and* K_σ^{hid}.

7. *A smooth embedding*

$$\jmath_\sigma : X_\sigma \hookrightarrow \widehat{X}_\sigma$$

and a smooth zero-response shell realization

$$\chi_\sigma : \mathcal{I}_\sigma^{\mathrm{for}} \hookrightarrow \mathcal{G}_\sigma^{\mathrm{sel}}$$

whose image is exactly

$$\mathcal{B}_\sigma^{\mathrm{sel}} \cap \mathcal{Z}_\sigma^0 \cap \left(\Lambda_\sigma^{\mathrm{sel,sh}}\right)^{-1}(0),$$

and such that

$$\Lambda_\sigma^{\mathrm{sel,dat}}\big(\chi_\sigma(w)\big) = \big(\jmath_\sigma(\mathcal{R}_\sigma(w)), \mathfrak{C}_\sigma^{\mathrm{cur}}(w), \mathfrak{C}_\sigma^{\mathrm{eq}}(w)\big) \tag{11}$$

for every $w \in \mathcal{I}_\sigma^{\mathrm{for}}$.

8. *Fixed-data shell solvability on the selector-zero locus:*

$$\Lambda_\sigma^{\mathrm{sel,dat}}\big(\mathcal{B}_\sigma^{\mathrm{sel}} \cap \mathcal{Z}_\sigma^0\big) = \Lambda_\sigma^{\mathrm{sel,dat}}\Big(\mathcal{B}_\sigma^{\mathrm{sel}} \cap \mathcal{Z}_\sigma^0 \cap \left(\Lambda_\sigma^{\mathrm{sel,sh}}\right)^{-1}(0)\Big). \tag{12}$$

9. *The inert zero-hidden kernel*

$$K_\sigma^0 \equiv \ker\big(D\Lambda_\sigma^{\mathrm{sel,dat}}\big|_{V_\sigma^0}\big) \cap \ker\big(D\Lambda_\sigma^{\mathrm{sel,sh}}\big|_{V_\sigma^0}\big)$$

and the orthogonal projector

$$P_\sigma^{\mathrm{co}} : V_\sigma^0 \to K_\sigma^{0\perp} \cap V_\sigma^0.$$

10. *Selector-law coercivity on data-invisible directions: there exists* $\kappa_\sigma^{\mathrm{sel}} > 0$ *such that for every*

$$w \in \mathcal{I}_\sigma^{\mathrm{for}}, \qquad \delta g \in T_{\chi_\sigma(w)}\mathcal{G}_\sigma^{\mathrm{sel}}$$

satisfying

$$D\Lambda_\sigma^{\mathrm{sel,dat}} \, \delta g = 0,$$

one has

$$\|D\Lambda_\sigma^{\mathrm{sel,sh}}\,\delta g\|^2 + \left\langle \Sigma_\sigma^{\mathrm{hid}}\delta g, K_\sigma^{\mathrm{hid}}\Sigma_\sigma^{\mathrm{hid}}\delta g\right\rangle \geq \kappa_\sigma^{\mathrm{sel}}\left(\|P_\sigma^{\mathrm{co}}((I-\Sigma_\sigma^{\mathrm{hid}})\delta g)\|^2 + \|\Sigma_\sigma^{\mathrm{hid}}\delta g\|^2\right). \tag{13}$$

Remark 3. *Definition 3 is deeper than the current combined response package in exactly the way now needed. It still carries a selector-zero locus with an inert zero-hidden kernel. The combined response law must therefore be recovered as the faithful zero-hidden quotient together with the explicit hidden response factor.*

5. Canonical Combined Response Law Induced by the Selector Law

Definition 4 (Canonical combined response law induced by the selector law)**.** *Assume Definition 3. Choose once and for all a reference shell point*

$$w_\sigma^* \in \mathcal{I}_\sigma^{\mathrm{for}}, \qquad q_\sigma^* \equiv \chi_\sigma(w_\sigma^*).$$

Define the canonical combined response substrate law as follows.

1. *The faithful zero-hidden quotient space is*

$$\mathcal{Q}_\sigma \equiv K_\sigma^{0\perp} \cap V_\sigma^0.$$

2. *The faithful zero-hidden coordinate is the affine map*

$$\pi_\sigma^{\mathrm{co}}(z) \equiv P_\sigma^{\mathrm{co}}(z - q_\sigma^*), \qquad z \in \mathcal{Z}_\sigma^0. \tag{14}$$

3. *The zero-hidden quotient body is*

$$\mathcal{Q}_\sigma^{\mathrm{bod}} \equiv \left\{\pi_\sigma^{\mathrm{co}}(z) : z \in \mathcal{B}_\sigma^{\mathrm{sel}} \cap \mathcal{Z}_\sigma^0\right\}. \tag{15}$$

4. *The primitive substrate space is*

$$\mathcal{M}_\sigma \equiv \mathcal{Q}_\sigma \oplus H_\sigma, \tag{16}$$

and the primitive substrate body is the product body

$$\mathcal{P}_\sigma \equiv \mathcal{Q}_\sigma^{\mathrm{bod}} \times D_\sigma. \tag{17}$$

5. *The affine combined response map is*

$$\Xi_\sigma(m,\eta) \equiv (\Lambda_\sigma^{\mathrm{sel,dat}}(q_\sigma^* + m), \Lambda_\sigma^{\mathrm{sel,sh}}(q_\sigma^* + m), \eta) \tag{18}$$

for $(m,\eta) \in \mathcal{M}_\sigma$, with components

$$\Xi_\sigma^{\mathrm{dat}}(m,\eta) \equiv \Lambda_\sigma^{\mathrm{sel,dat}}(q_\sigma^* + m), \qquad \Xi_\sigma^{\mathrm{sh}}(m,\eta) \equiv \Lambda_\sigma^{\mathrm{sel,sh}}(q_\sigma^* + m), \qquad \Xi_\sigma^{\mathrm{hid}}(m,\eta) \equiv \eta.$$

6. *The compact symmetry group and time reversal are the descended selector symmetries:*

$$\mathbb{H}_\sigma \equiv \{(g_{\mathrm{dat}}, g_{\mathrm{sh}}, g_{\mathrm{hid}}) : (g_{\mathrm{sel}}, g_{\mathrm{dat}}, g_{\mathrm{sh}}) \in \mathbb{S}_\sigma\},$$

where g_{hid} is the restriction of g_{sel} to H_σ, and

$$\rho_\sigma \equiv (\vartheta_\sigma^{\mathrm{sel}}|_{\mathcal{Y}_\sigma^{\mathrm{obs}}}, \vartheta_\sigma^{\mathrm{sel}}|_{E_\sigma^{\mathrm{amb}}}, \vartheta_\sigma^{\mathrm{sel}}|_{H_\sigma}).$$

7. The exact shell realization map is

$$\zeta_\sigma(w) \equiv (\pi_\sigma^{\mathrm{co}}(\chi_\sigma(w)), 0), \qquad w \in \mathcal{I}_\sigma^{\mathrm{for}}. \tag{19}$$

Remark 4. *The choice of reference shell point q_σ^* fixes only the affine origin of the faithful zero-hidden coordinate. Changing it translates the quotient coordinate on the same faithful zero-hidden quotient and does not alter any benchmark-relevant constituent of the induced combined response law.*

Proposition 1 (The selector law induces a combined response substrate law)**.** *Assume Definitions 3, 2, and 4. Then the canonical data of Definition 4 satisfy every requirement of Definition 2.*

Proof. Because $K_\sigma^0 \subset V_\sigma^0$ and $\mathcal{Q}_\sigma = K_\sigma^{0\perp} \cap V_\sigma^0$, every $z \in \mathcal{Z}_\sigma^0$ can be written uniquely as

$$z = q_\sigma^* + m + k \qquad \text{with } m \in \mathcal{Q}_\sigma, \quad k \in K_\sigma^0.$$

Since $\Lambda_\sigma^{\mathrm{sel,dat}}$ and $\Lambda_\sigma^{\mathrm{sel,sh}}$ are affine and K_σ^0 lies in the kernels of their linear parts on V_σ^0, the definitions in (18) are well defined and affine on $\mathcal{Q}_\sigma$.

The set $\mathcal{B}_\sigma^{\mathrm{sel}} \cap \mathcal{Z}_\sigma^0$ is compact and convex, so its image $\mathcal{Q}_\sigma^{\mathrm{bod}}$ under the affine map π_σ^{co} is compact and convex. Because $\mathcal{B}_\sigma^{\mathrm{sel}}$ has nonempty interior and splits exactly as in (9) with $0 \in \operatorname{int} D_\sigma$, the selector-zero slice carries nonempty relative interior in $\mathcal{B}_\sigma^{\mathrm{sel}}$. Therefore $\mathcal{Q}_\sigma^{\mathrm{bod}}$ has nonempty interior in $\mathcal{Q}_\sigma$. It follows that the product body $\mathcal{P}_\sigma = \mathcal{Q}_\sigma^{\mathrm{bod}} \times D_\sigma$ is compact, convex, and has nonempty interior in $\mathcal{M}_\sigma = \mathcal{Q}_\sigma \oplus H_\sigma$.

The response image is

$$\mathcal{B}_\sigma^{\mathrm{resp}} = \left\{ (\Lambda_\sigma^{\mathrm{sel,dat}}(q_\sigma^* + m), \Lambda_\sigma^{\mathrm{sel,sh}}(q_\sigma^* + m), \eta) : m \in \mathcal{Q}_\sigma^{\mathrm{bod}},\ \eta \in D_\sigma \right\},$$

so it is compact and convex. Its shell slice is

$$\left\{ (\Lambda_\sigma^{\mathrm{sel,dat}}(q_\sigma^* + m), \Lambda_\sigma^{\mathrm{sel,sh}}(q_\sigma^* + m), 0) : m \in \mathcal{Q}_\sigma^{\mathrm{bod}} \right\},$$

which has nonempty relative interior because $\mathcal{Q}_\sigma^{\mathrm{bod}}$ does. The exact response-image factorization (4) holds by construction from the product body (17).

Equation (5) is exactly (10). The descended symmetry and time-reversal data preserve $\mathcal{B}_\sigma^{\mathrm{resp}}$, the hidden-zero slice, the zero shell value, and K_σ^{hid} because the selector symmetries preserve $\Lambda_\sigma^{\mathrm{sel,dat}}$, $\Lambda_\sigma^{\mathrm{sel,sh}}$, $\Sigma_\sigma^{\mathrm{hid}}$, D_σ, and K_σ^{hid}.

For the shell realization, $\zeta_\sigma(\mathcal{I}_\sigma^{\mathrm{for}}) \subset \mathcal{P}_\sigma$ because $\chi_\sigma(\mathcal{I}_\sigma^{\mathrm{for}}) \subset \mathcal{B}_\sigma^{\mathrm{sel}} \cap \mathcal{Z}_\sigma^0$. Also

$$\Xi_\sigma^{\mathrm{sh}}(\zeta_\sigma(w)) = \Lambda_\sigma^{\mathrm{sel,sh}}(\chi_\sigma(w)) = 0, \qquad \Xi_\sigma^{\mathrm{hid}}(\zeta_\sigma(w)) = 0.$$

Conversely, if

$$(m, 0) \in \mathcal{P}_\sigma \cap (\Xi_\sigma^{\mathrm{sh}})^{-1}(0) \cap (\Xi_\sigma^{\mathrm{hid}})^{-1}(0),$$

then there exists

$$z \in \mathcal{B}_\sigma^{\mathrm{sel}} \cap \mathcal{Z}_\sigma^0$$

with $m = \pi_\sigma^{\mathrm{co}}(z)$. Since

$$\Xi_\sigma^{\mathrm{sh}}(m, 0) = \Lambda_\sigma^{\mathrm{sel,sh}}(z) = 0,$$

we have

$$z \in \mathcal{B}_\sigma^{\mathrm{sel}} \cap \mathcal{Z}_\sigma^0 \cap \big(\Lambda_\sigma^{\mathrm{sel,sh}}\big)^{-1}(0).$$

Because χ_σ is a diffeomorphism onto that exact selector-zero shell-zero locus, there is a unique $w \in \mathcal{I}_\sigma^{\mathrm{for}}$ with $z = \chi_\sigma(w)$, and therefore $(m, 0) = \zeta_\sigma(w)$. Hence ζ_σ has exactly the required image.

Now consider the map

$$\Phi_\sigma : \mathcal{Q}_\sigma^{\mathrm{bod}} \cap \big(\Xi_\sigma^{\mathrm{sh}}\big)^{-1}(0) \to \mathcal{B}_\sigma^{\mathrm{resp}} \cap \big(\mathcal{Y}_\sigma^{\mathrm{obs}} \oplus \{0\} \oplus \{0\}\big), \qquad \Phi_\sigma(m) \equiv \big(\Xi_\sigma^{\mathrm{dat}}(m, 0), 0, 0\big).$$

This map is surjective by construction. It is injective because if

$$\Phi_\sigma(m_1) = \Phi_\sigma(m_2),$$

then

$$\Xi_\sigma^{\mathrm{dat}}(m_1, 0) = \Xi_\sigma^{\mathrm{dat}}(m_2, 0), \qquad \Xi_\sigma^{\mathrm{sh}}(m_1, 0) = \Xi_\sigma^{\mathrm{sh}}(m_2, 0) = 0.$$

Therefore $m_1 - m_2 \in \mathcal{Q}_\sigma$ lies in the kernels of the linear parts of both descended affine maps, hence in

$$\ker\big(D\Lambda_\sigma^{\mathrm{sel,dat}}\big|_{V_\sigma^0}\big) \cap \ker\big(D\Lambda_\sigma^{\mathrm{sel,sh}}\big|_{V_\sigma^0}\big) \cap \mathcal{Q}_\sigma = K_\sigma^0 \cap K_\sigma^{0\perp} = \{0\}.$$

So $m_1 = m_2$, and Φ_σ is a diffeomorphism. Since ζ_σ is a diffeomorphism onto the shell-zero hidden-zero slice, the composition $\Xi_\sigma \circ \zeta_\sigma$ is a diffeomorphism onto

$$\mathcal{B}_\sigma^{\mathrm{resp}} \cap \big(\mathcal{Y}_\sigma^{\mathrm{obs}} \oplus \{0\} \oplus \{0\}\big).$$

For the data compatibility (6), the vectors

$$\chi_\sigma(w) - q_\sigma^* - \pi_\sigma^{\mathrm{co}}\big(\chi_\sigma(w)\big)$$

lie in K_σ^0, so they are invisible to the affine readouts on V_σ^0. Hence

$$\Xi_\sigma^{\mathrm{dat}}\big(\zeta_\sigma(w)\big) = \Lambda_\sigma^{\mathrm{sel,dat}}\big(\chi_\sigma(w)\big) = \big(\jmath_\sigma(\mathcal{R}_\sigma(w)), \mathfrak{C}_\sigma^{\mathrm{cur}}(w), \mathfrak{C}_\sigma^{\mathrm{eq}}(w)\big).$$

For fixed-data solvability, let

$$u \in \mathrm{pr}_{\mathcal{Y}_\sigma^{\mathrm{obs}}}\big(\mathcal{B}_\sigma^{\mathrm{resp}}\big).$$

Then $u = \Lambda_\sigma^{\mathrm{sel,dat}}(z)$ for some

$$z \in \mathcal{B}_\sigma^{\mathrm{sel}} \cap \mathcal{Z}_\sigma^0.$$

By (12), there exists

$$z_0 \in \mathcal{B}_\sigma^{\mathrm{sel}} \cap \mathcal{Z}_\sigma^0 \cap \big(\Lambda_\sigma^{\mathrm{sel,sh}}\big)^{-1}(0)$$

with $\Lambda_\sigma^{\mathrm{sel,dat}}(z_0) = u$. Setting

$$m_0 \equiv \pi_\sigma^{\mathrm{co}}(z_0)$$

gives

$$(m_0, 0) \in \mathcal{P}_\sigma \cap \left(\Xi_\sigma^{\mathrm{sh}}\right)^{-1}(0) \cap \left(\Xi_\sigma^{\mathrm{hid}}\right)^{-1}(0)$$

and

$$\Xi_\sigma^{\mathrm{dat}}(m_0, 0) = u,$$

which proves (7).

Finally, let $w \in \mathcal{I}_\sigma^{\mathrm{for}}$ and let

$$\delta m = (\delta q, \delta\eta) \in T_{\zeta_\sigma(w)}\mathcal{M}_\sigma = \mathcal{Q}_\sigma \oplus H_\sigma$$

satisfy

$$D\Xi_\sigma^{\mathrm{dat}}\big(\zeta_\sigma(w)\big)\,\delta m = 0.$$

Then

$$D\Lambda_\sigma^{\mathrm{sel,dat}}\,\delta q = 0$$

with $\delta q \in \mathcal{Q}_\sigma \subset V_\sigma^0$. Applying (13) to $\delta g = \delta q$ gives

$$\|D\Lambda_\sigma^{\mathrm{sel,sh}}\,\delta q\|^2 \geq \kappa_\sigma^{\mathrm{sel}}\,\|\delta q\|^2,$$

because $\Sigma_\sigma^{\mathrm{hid}}\delta q = 0$ and $P_\sigma^{\mathrm{co}}\delta q = \delta q$. Therefore

$$\begin{aligned}\left\|D\Xi_\sigma^{\mathrm{sh}}\big(\zeta_\sigma(w)\big)\,\delta m\right\|^2 + \left\langle D\Xi_\sigma^{\mathrm{hid}}\big(\zeta_\sigma(w)\big)\,\delta m, K_\sigma^{\mathrm{hid}} D\Xi_\sigma^{\mathrm{hid}}\big(\zeta_\sigma(w)\big)\,\delta m\right\rangle &= \|D\Lambda_\sigma^{\mathrm{sel,sh}}\,\delta q\|^2 + \left\langle \delta\eta, K_\sigma^{\mathrm{hid}}\delta\eta\right\rangle \\ &\geq \kappa_\sigma^{\mathrm{sel}}\|\delta q\|^2 + \kappa_\sigma^{\mathrm{hid}}\|\delta\eta\|^2 \\ &\geq \kappa_\sigma^{\mathrm{resp}}\big(\|\delta q\|^2 + \|\delta\eta\|^2\big),\end{aligned}$$

where

$$\kappa_\sigma^{\mathrm{resp}} \equiv \min\{\kappa_\sigma^{\mathrm{sel}}, \kappa_\sigma^{\mathrm{hid}}\} > 0.$$

This is exactly (8). All requirements of Definition 2 are satisfied. □

6. Forced Constituents and Uniqueness

Proposition 2 (All combined-response constituents are forced by the selector law)**.** *Assume Definition 3. Then the following constituents are fixed by the selector law itself.*

1. *The primitive substrate state is necessarily the faithful quotient of the selector-zero translation space by the inert zero-hidden kernel, represented canonically by*

$$\mathcal{Q}_\sigma = K_\sigma^{0\perp} \cap V_\sigma^0.$$

2. *The primitive substrate body is necessarily the product $\mathcal{P}_\sigma = \mathcal{Q}_\sigma^{\mathrm{bod}} \times D_\sigma$, where $\mathcal{Q}_\sigma^{\mathrm{bod}}$ is the quotient image* (15) *of the selector-zero body.*

3. *The affine observer-data and shell-response components of the combined response map are necessarily the descended selector readouts of* (18) *on the faithful zero-hidden quotient, and the hidden response component is necessarily the explicit hidden coordinate in H_σ.*

4. *The response-image hidden factor is necessarily D_σ, and the positive hidden-sector operator is*

necessarily K_σ^{hid}.

5. *The descended symmetry / time-reversal structure, the exact zero-response shell realization, fixed-data shell solvability, and the response-kernel coercivity mechanism are all induced by the same selector law and are not additional combined-response freedom.*

Proof. Item (1) is exactly the faithful zero-hidden quotient singled out by the selector law: once the selector-zero locus and the inert kernel are fixed, the only faithful Euclidean zero-hidden coordinates are those that quotient out K_σ^0, represented canonically by $K_\sigma^{0\perp} \cap V_\sigma^0$.

Item (2) is then forced because the combined response law must retain precisely the selector-zero quotient body together with the already fixed hidden selector body. There is no second compatible hidden factor once D_σ is fixed, and there is no second compatible faithful zero-hidden body once the quotient coordinate (14) is fixed.

Item (3) follows because $\Lambda_\sigma^{\mathrm{sel,dat}}$ and $\Lambda_\sigma^{\mathrm{sel,sh}}$ are affine and the inert kernel lies in the kernels of their linear parts on V_σ^0. Therefore those readouts descend uniquely to the faithful quotient, while the hidden response must be exactly the explicit hidden coordinate if the factorization (4) is to hold with hidden body D_σ.

Item (4) is the hidden package already present in Definition 3. Item (5) is the content of Proposition 1: the descended reversible structure, the exact zero-response shell realization, fixed-data solvability, and the coercivity inequality all come directly from the same selector package and carry no extra independent freedom at combined-response level. □

Theorem 1 (Combined response substrate law necessity / uniqueness from selector law)**.** *Assume Definition 3. Then:*

1. *the canonical construction of Definition 4 yields a combined response substrate law in the sense of Definition 2;*

2. *every combined-response constituent currently left open by the active frontier is forced by the selector law: the faithful primitive substrate state, the primitive substrate body, the affine combined response map, the response-image factorization into shell and hidden sectors, the positive hidden-sector operator, the descended symmetry and time-reversal structure, the exact zero-response shell realization, fixed-data shell solvability, and the response-kernel coercivity mechanism;*

3. *any other combined response law compatible with the same selector law differs from the canonical one only by the harmless affine choice of faithful zero-hidden origin and by the obvious orthogonal coordinate identifications on* $\mathcal{Q}_\sigma$ *and* H_σ *preserving the recorded data and shell coordinates;*

4. *consequently the combined response substrate law is not an independent remaining freedom once the selector law is fixed.*

Proof. Item (1) is Proposition 1. Item (2) is Proposition 2. For item (3), changing the reference shell point q_σ^* only translates the affine coordinate on the same faithful zero-hidden quotient, while changing orthogonal coordinates on $\mathcal{Q}_\sigma$ or H_σ only produces the obvious coordinate identifications preserving the same descended selector data, shell readout, and hidden quadratic form. No additional compatible freedom remains. Item (4) is the routing consequence of the previous items. □

7. Benchmark Preservation

Theorem 2 (The derived combined response law preserves the same one-metric benchmark package)**.** *Assume Definition 3. Then the benchmark output statements of Definition 1 remain unchanged:*

$$\mathscr{G}_{\mathrm{out}}(\mathbf{w}) = g^{\sharp}, \qquad \mathscr{T}_{\mathrm{out}}(\mathbf{w}, \psi) = T^{\mathrm{matter}}_{\mu\nu}[g^{\sharp}, \psi]$$

for every $\mathbf{w} \in \mathcal{S}_{\mathrm{prim}}$ *and every matter configuration* $\psi \in \Psi$*. Consequently the deeper selector law and the induced combined response law introduce neither a second operative metric nor an enlarged low-energy matter law.*

Proof. The present note changes only the upstream origin of the combined response package. It does not alter the primitive shell $\mathcal{S}_{\mathrm{prim}}$, the recorded metric map $\mathscr{G}_{\mathrm{out}}$, or the recorded matter map $\mathscr{T}_{\mathrm{out}}$. Those remain the fixed continuation data of Definition 1, so (2) continues to hold exactly. □

8. Main Theorem

Theorem 3 (The remaining burden moves below the combined-response layer)**.** *Assume the fixed primitive continuation data of Definition 1 and, for each sector* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*, a zero-hidden selector generator law in the sense of Definition 3. Then:*

1. *the combined response substrate law of Definition 2 is canonically induced by the selector law after faithful inert-kernel reduction;*
2. *the faithful primitive substrate state, the primitive substrate body, the affine combined response map, the response-image factorization into shell and hidden sectors, the positive hidden-sector operator, the descended symmetry and time-reversal structure, the exact zero-response shell realization, fixed-data shell solvability, and the response-kernel coercivity mechanism are all forced by that same selector law;*
3. *the one operative metric remains exactly* $g^{\sharp}$*, the ordinary matter route remains exactly the standard one through that metric, there is no second operative metric, and the low-energy matter law is not enlarged;*
4. *consequently the benchmark-facing burden after the current ambient-dynamics frontier is discharged in the stronger form now required on the active line: the combined response substrate law is no longer an unexplained stopping point, but a canonical and necessary reduction of a deeper zero-hidden selector generator law.*

Proof. Item (1) is Theorem 1(1). Item (2) is Theorem 1(2). Item (3) is Theorem 2. Item (4) is the combined routing consequence. □

Corollary 1 (What remains open after this theorem)**.** *After Theorem 3, the remaining honest burden is no longer to justify the combined response substrate law itself on the active primitive-observer-reduction line. The next honest burden is deeper again: first try to derive or uniquely force the zero-hidden selector generator law from still more primitive explicit substrate structure in a way that actually changes benchmark-facing status; if that cannot be done cleanly, the admissible fallback is a genuinely smaller theorem-level collapse, sharper necessity result, or sharper uniqueness result strictly inside that selector law while preserving the same benchmark one-metric package and claim boundary.*

Remark 5. *Theorem 3 converts the combined response substrate law from the current live burden into a canonical reduction of a deeper selector law. What remains open is why that selector law itself should hold, rather than becoming the next disciplined stopping point. Another ambient-package restatement, trace-core restatement, completion-core restatement, exact-shell affine nucleus restatement, or same-law combined-response relay would therefore leave the current live burden unchanged.*

9. Conclusion

The positive result of this note is a direct forcing theorem for the current combined-response stopping point. The previous ambient-dynamics theorem had already shown that the completion-balance ambient package is a canonical reduction of the deeper combined response substrate law. The present note spends the first honest move available there: it shows that the combined response law is itself canonically induced by a deeper zero-hidden selector generator law after quotienting out exactly the selector-zero directions already invisible to the recorded data and shell readouts.

That gain directly addresses the precise constituents that had become the live burden: the primitive substrate state, the primitive substrate body, the affine combined response map, the response-image factorization, the hidden-sector operator, the reversible structure, the exact zero-response shell realization, fixed-data shell solvability, and the response-kernel coercive control. Within the selector-law class those are no longer optional inserted ingredients. They are forced outputs, unique up to the harmless affine choice of faithful zero-hidden origin and the obvious orthogonal identifications on the faithful quotient and hidden factor.

The benchmark boundary remains fixed throughout. The same primitive shell remains the shell carrying the observer outputs, so the one operative metric is still exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, and the low-energy matter law is not enlarged. This is therefore a disciplined upstream derivational gain, not a final first-principles completion claim.

The next honest burden is now one level deeper again. The note does not yet derive the zero-hidden selector generator law from ultimate substrate microphysics. The next benchmark-facing attempt should therefore first try to derive or uniquely force that selector law from still more primitive explicit substrate structure in a way that actually changes benchmark-facing status. If that cannot be done cleanly, the admissible fallback is a genuinely smaller theorem-level collapse, sharper necessity result, or sharper uniqueness result strictly inside that selector law. Another ambient-package restatement, trace-core restatement, completion-core restatement, exact-shell affine nucleus restatement, or another same-law combined-response relay that preserves the same selector-law stopping point would not meet that burden, and no stronger first-principles promotion language is licensed yet.